COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	10/09/2026	**DATE SUBMITTED:**	10/09/2026
Group	2	**INSTRUCTOR:**	ENGR. JIMLORD M. QUEJADO

ACTIVITY NO. 11.1 PLANNING

1. PROCEDURE OUTPUT

What procedures did you perform?
Write down your 3 proposed systems to develop:

Pickleball Court Booking & Management System	- Reserve courts, check availability, manage schedule, maintain waitlist, record fees and payments - Prevents double-booking, eliminates manual log errors, provides organized chronological schedules - Target Users: sports club admins, coaches, players
Internet Cafe / Computer Rental Time & Billing System	- Automates PC timers, prepaid hours, user accounts, and daily income reports. - Solves unrecorded usage time, manual billing errors, and unnoticed computer downtime. - Target users: shop owners, attendants, students
Boarding House Rent & Payment Tracker Description	- Records tenant information, room details, monthly rent amount, due dates, payments made, remaining balance, and generates income reports - Automatically calculates balances, flags overdue accounts, and tracks room occupancy status - Solves: Lost paper logs, forgotten payments, manual computation errors, unclear who has paid or not - Target Users: Landlords, boarding house managers, tenants

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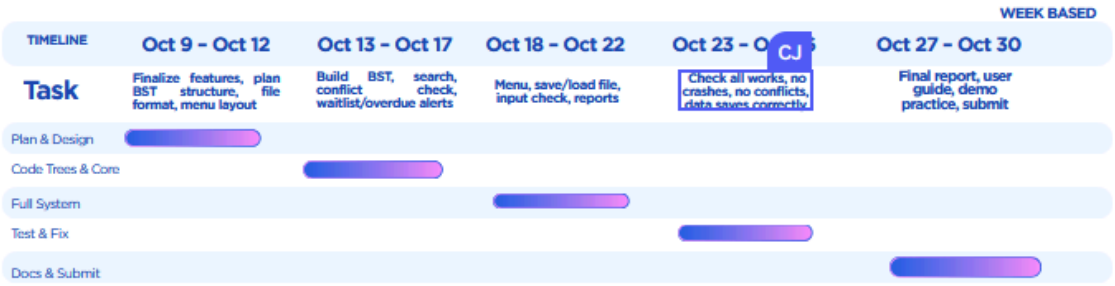
SDLC: We chose Waterfall because it matches exactly what we learned in our DSA topics. All three systems use Trees like BST as their main structure, and we need to plan that structure first before coding. Waterfall follows the

same step-by-step order — plan, design, code, test — just like how we study Arrays, Linked Lists, Stacks, Queues, Searching, Sorting, and Trees one by one. It fits our Pickleball, Internet Cafe, and Boarding House systems perfectly and keeps everything simple and organized. That is why Waterfall is the right choice for our project.

Pickleball Court Booking & Management System	Internet Cafe / Computer Rental Time & Billing System	Boarding House Rent & Payment Tracker Description
1. What are your operating hours and how many courts do you have? 2. How do you record or schedule bookings right now? 3. What details do you collect from players who book? 4. How do you make sure no one books the same court and time? 5. How much do you charge and how do you record payments? 6. What happens if someone cancels or does not show up? 7. If all courts are full, how do you manage people waiting? 8. Who is allowed to approve or change bookings? 9. What reports do you need for daily bookings, earnings, and peak hours? 10. What is the hardest or most confusing part of your current process?	1. How many computers do you have, and how much is the rate per hour? 2. How do you check if a PC is available or being used? 3. What information do you get from every user? 4. How do you calculate the total time and payment? 5. Do you offer prepaid load, packages, or discounts? 6. How do you manage the waiting list when all PCs are full? 7. How do you track daily earnings and most-used units? 8. Who can change rates, add time, or correct records? 9. How do you handle time extensions in the middle of a session? 10. What problems happen often with your current way of recording?	1. How many rooms or bed spaces do you manage, and what are the monthly rental rates for each? 2. How do you record tenant personal information, contact details, and emergency contacts? 3. What is your process for tracking contract start dates, lease durations, and due dates? 4. How do you check which rooms are currently occupied, reserved, or vacant? 5. How do you record rental payments, partial payments, and remaining balances? 6. Do you charge late fees or penalties for overdue rent, and how are they calculated? 7. How are utility expenses (electricity, water, Wi-Fi) recorded or divided among tenants? 8. How do you manage the waitlist or reservation list when all rooms/beds are fully occupied?

		9. What happens when a tenant moves out, cancels early, or requests a security deposit refund? 10. What is the most confusing or time-consuming part of your current paper-based or manual log recording?

GANTT CHART



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **Grammarly**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar correction**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

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Extent of Use: **Did not replace the original thinking**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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